

ORC · STANDALONE NOTE 3

On the Direct Detection of a Single-Atom Gravitational Field

A Feasibility Analysis

*Newtonian field 10^{-19} m/s² at 1 nm · Integration time 10^{33} s (longer than universe age) · 22.7 OOM gap from sota
Even if achievable, would not test quantum gravity — BMV protocols remain the proper target.*

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On the question: can a single atom be felt by another body through gravity alone?

It can — at 1 nm, with acceleration 10^{-19} m/s². The integration time to detect this exceeds the age of the universe by 23 orders of magnitude. The numbers, again, are the answer.

ABSTRACT

We analyze the feasibility of directly detecting the gravitational field of a single atom using state-of-the-art precision force sensors. For a hydrogen atom ($m_H = 1.67 \times 10^{-27}$ kg), the Newtonian gravitational acceleration at distance $d = 1$ nm is $g(d) = Gm_H/d^2 \approx 1.1 \times 10^{-19}$ m/s². The corresponding force on a 100 nm levitated test sphere (mass 10^{-18} kg) is $F_{\text{grav}} \approx 1.1 \times 10^{-37}$ N. State-of-the-art force sensitivity for single levitated nanoparticles is $\sim 10^{-20}$ N/√Hz [1,2,3]. The integration time required to detect the single-atom signal at unity SNR is $t \sim 10^{33}$ s, many orders of magnitude longer than the age of the universe ($\sim 4 \times 10^{17}$ s). The current state-of-the-art tabletop gravity measurement [4] used 90 mg source masses — a gap of $\sim 10^{23}$ atoms (22.7 orders of magnitude) from the single-atom limit. Beyond this fundamental sensitivity gap, the experiment is dominated by electromagnetic backgrounds: the Casimir-Polder force from the same source atom exceeds its gravitational force by 10^{20} – 10^{26} at every relevant distance, and patch potentials from nearby surfaces produce forces 10^{28} times larger than the gravitational signal. **We conclude that direct detection of a single-atom gravitational field is not achievable with any foreseeable technology.** We further note that even if it were, it would not constitute a test of quantum gravity: a stationary atom produces the same Newtonian field as its classical mass distribution. Genuine quantum-gravity tests require coherent superpositions of position (the BMV protocol [5,6]), for which the relevant phase $\phi = Gm^2t/(\hbar d)$ is marginally accessible with $m \sim 10^{-14}$ kg, $d \sim 200$ μm, $t \sim 1$ s, giving $\phi \sim 0.32$ rad. We recommend the BMV target as the proper experimental program; this note documents why the single-atom-field framing is the wrong target.

1. Introduction and Motivation

The question 'can the gravitational pull of a single atom be felt' has a tempting structure: it appears to ask whether quantum mechanics and general relativity can be probed at their common boundary. The implicit reasoning is that if force-sensing technology continues to

improve, eventually even a single atomic mass will produce a detectable gravitational signature. We show in this note that this reasoning fails by extraordinary margins, and that the question itself — even if answered affirmatively — would not address the physics it appears to address.

The state-of-the-art tabletop gravity experiment [4] reported gravitational coupling between two 90 mg masses at sub-millimeter separations, a result that took multiple decades of technological development to achieve. Pushing beyond this — to milligram, microgram, nanogram, and ultimately single-atom source masses — would require sensitivity improvements that this note will quantify. We also examine the noise budget that any such experiment would need to overcome, focusing on electromagnetic backgrounds (Casimir-Polder, patch potentials, blackbody radiation pressure) that scale unfavorably with respect to the gravitational signal as source masses shrink.

Finally, we address a conceptual point that is often elided in this discussion: **direct detection of a single-atom gravitational field would not test quantum gravity**. A stationary atom produces a static Newtonian field $G \cdot m/r^2$ that is independent of whether gravity is fundamentally classical or quantum. The genuine test — whether gravitational interaction can mediate entanglement between two superposed masses — requires the BMV protocol [5,6], which operates at much larger mass scales. We treat this distinction with care because the single-atom-detection framing has appeared informally in popular discussions and merits explicit demarcation.

2. The Newtonian Field of a Single Atom

For a hydrogen atom of mass $m_H = 1.6735 \times 10^{-27}$ kg, the Newtonian gravitational acceleration at distance d is

$$g(d) = G m_H / d^2.$$

Distance d	$g(d)$ [m/s ²]	Comparison
1 nm	1.12×10^{-19}	Atomic-scale separation
10 nm	1.12×10^{-21}	Surface-physics scale
100 nm	1.12×10^{-23}	Optical wavelength scale
1 μ m	1.12×10^{-25}	Cell-size scale
10 μ m	1.12×10^{-27}	Membrane-thickness scale
1 mm	1.12×10^{-31}	Tabletop scale
1 m	1.12×10^{-37}	Lab scale

For comparison, Earth's surface gravity is 9.81 m/s² and the smallest gravitational acceleration ever measured in the laboratory is on the order of 10^{-13} m/s² (matter-wave interferometry, atom-fountain gradiometers). The single-atom field at any achievable separation lies many orders of magnitude below this floor.

3. Comparison to State-of-the-Art Sensitivity

The most sensitive single-particle force detectors currently in operation are levitated nanoparticle systems pioneered by the Geraci, Aspelmeyer, Moore, and Novotny groups [1,2,3]. Reported force sensitivities approach 10^{-20} N/ $\sqrt{\text{Hz}}$ for single particles at cryogenic temperatures with feedback cooling.

The gravitational force on such a test mass (100 nm silica sphere, $m_{\text{test}} \approx 10^{-18}$ kg) from a single H atom at distance d is $F_{\text{grav}} = G m_H m_{\text{test}} / d^2$. Integration time to detect at unity signal-to-noise scales as $t = (S_F / F_{\text{grav}})^2$:

d	F_grav [N]	Integration time t	Comparison to age of universe
1 nm	1.1×10^{-37}	8.0×10^{33} s	$6 \times 10^{15} \times$ age of universe
10 nm	1.1×10^{-39}	8.0×10^{37} s	$6 \times 10^{19} \times$ age of universe
100 nm	1.1×10^{-41}	8.0×10^{41} s	$6 \times 10^{23} \times$ age of universe
1 μm	1.1×10^{-43}	8.0×10^{45} s	$6 \times 10^{27} \times$ age of universe

Even at 1 nm separation — closer than the test sphere's own surface contact would allow — the integration time exceeds the age of the universe by sixteen orders of magnitude. Improving sensitivity by another four orders of magnitude ($S_F \sim 10^{-24}$ N/ $\sqrt{\text{Hz}}$, far beyond any anticipated technology) reduces this to ten orders of magnitude longer than the universe age. No reasonable extrapolation of current technology bridges this gap.

4. The Gap to Current Sota Tabletop Gravity

Westphal et al. [4] reported in 2021 the smallest source mass for which gravitational coupling has been directly detected: 90 mg, observed at sub-millimeter separation through coupling to a torsion pendulum. This is the current frontier of small-mass gravity detection. The gap to single-atom detection is

$$\Delta = m_{\text{Westphal}} / m_H = 9 \times 10^{-5} \text{ kg} / 1.67 \times 10^{-27} \text{ kg} \approx 5.4 \times 10^{22} \text{ atoms},$$

or 22.7 orders of magnitude. To bridge this gap, source mass would need to shrink by a factor of 10^{23} . There is no demonstrated technology path for this; even ambitious next-generation experiments (mg-scale sources at sub- μm separation, using levitated particles as both source and probe) project source-mass reductions of at most 10^5 – 10^6 over the next decade.

5. Why Sensitivity Alone Is Not the Issue: The EM Background

Even granting indefinite improvement in pure force sensitivity, the experiment is foreclosed by electromagnetic backgrounds from the same source atom. We tabulate the dominant systematics:

5.1 Casimir-Polder Force from the Source Atom

A neutral atom interacts with a polarizable test body through the Casimir-Polder dispersion force. For an H atom (static polarizability $\alpha \approx 4.5 a_0^3$) and a dielectric test sphere at distance d in the retarded regime [7,8]:

$$F_{\text{CP}} \sim 3\hbar c \alpha / (8\pi \epsilon_0 d^5).$$

d	F_grav [N]	F_CP [N]	Ratio F_CP / F_grav
1 nm	1.1×10^{-37}	3.2×10^{-11}	2.8×10^{26}
10 nm	1.1×10^{-39}	3.2×10^{-16}	2.8×10^{23}
100 nm	1.1×10^{-41}	3.2×10^{-21}	2.8×10^{20}
1 μm	1.1×10^{-43}	3.2×10^{-26}	2.8×10^{17}

The Casimir-Polder force from a single source atom exceeds its gravitational force by 17 to 26 orders of magnitude across the relevant distance range. The gravitational signal is buried in an EM background that is generated by the source itself and cannot be screened away.

5.2 Patch Potentials

Any nearby surface (the housing, the sphere itself, electrodes) carries a patch-potential landscape with $V_{\text{patch}} \sim 100$ mV characteristic amplitudes [9]. The induced force on a 100 nm sphere is approximately $F_{\text{patch}} \sim q_{\text{eff}}^2 / (4\pi\epsilon_0 d^2)$ with $q_{\text{eff}} \sim V_{\text{patch}} \cdot 4\pi\epsilon_0 R_{\text{test}}$. Numerically, $F_{\text{patch}} \sim 10^{-13}$ N at 100 nm — exceeding single-atom gravity by approximately 28 orders of magnitude. No grounding scheme reduces this below the gravitational signal level.

5.3 Blackbody Radiation Pressure

At cryogenic operating temperature $T = 4$ K, blackbody radiation pressure on a 100 nm test sphere is $F_{\text{BB}} \sim 4 \times 10^{-28}$ N — eight orders of magnitude above single-atom gravity at 100 nm. In a perfectly symmetric thermal environment this force averages to zero, but residual thermal asymmetries and fluctuations contribute systematically.

6. Noise-Budget Summary

Effect	Magnitude at 100 nm	Status
Single-atom gravity	1.1×10^{-41} N	Target signal
Sota sensitivity floor	$\sim 10^{-20}$ N/ $\sqrt{\text{Hz}}$	21 OOM above signal
Casimir-Polder (source)	3.2×10^{-21} N	20 OOM above signal
Patch potentials	$\sim 10^{-13}$ N	28 OOM above signal
Blackbody (4 K)	$\sim 4 \times 10^{-28}$ N	13 OOM above signal
Time to bridge SNR=1	$\sim 10^{41}$ s	23 OOM > universe age

7. Why This Would Not Test Quantum Gravity

Even granting hypothetical detection of a single-atom gravitational field, the experiment would not constitute a test of quantum gravity. The reason is fundamental: a stationary atom in a position eigenstate produces a Newtonian field $G \cdot m/r^2$ that is identical to its classical counterpart. The quantum nature of gravity (whether gravity must be quantized, whether gravitational interactions can carry quantum information) does not enter the field of a stationary, classical-position source.

The genuine quantum-gravity test, proposed independently by Bose et al. [5] and Marletto and Vedral [6] (the BMV protocol), requires two masses each in coherent superposition of two positions. If gravitational interaction is fundamentally quantum, the two superposed mass states become entangled through their gravitational coupling — a result that has no classical analog. The phase accumulated through gravitational interaction in time t at separation d is

$$\phi = G m^2 t / (\hbar d).$$

For the parameters considered most accessible — $m \sim 10^{-14}$ kg (microgram-scale superpositions), $d \sim 200$ μm separation, $t \sim 1$ s coherence time — we obtain $\phi \approx 0.32$ rad. Detection requires $\phi > \sim 0.1$ rad, so the protocol is marginally accessible. Active experimental programs at Vienna (Aspelmeyer), at UCL, and elsewhere are developing the matter-wave interferometry, mass loading, and decoherence-suppression techniques required.

The relevant target for quantum-gravity tests is the BMV protocol with $m \sim 10^{-14}$ kg, not direct field detection from $m \sim 10^{-27}$ kg sources. The two questions differ by 13 orders of magnitude in source mass and by infinite epistemic distance: one is a search for a static Newtonian field, the other is a test of whether gravity carries quantum information.

8. Conclusions

Direct detection of a single-atom gravitational field is foreclosed by three independent barriers, each sufficient on its own. **First, sensitivity:** the integration time required to reach unity SNR exceeds the age of the universe by 16–27 orders of magnitude depending on separation, with no demonstrated technology path bridging the gap. **Second, electromagnetic backgrounds:** the Casimir-Polder force from the source atom itself exceeds its gravitational field by 17–26 orders of magnitude, and patch potentials from nearby surfaces add another 28 orders of magnitude. **Third, conceptual:** even if detection were possible, it would not test quantum gravity, since a stationary atom's field is identical to its classical Newtonian field.

The relevant experimental target for quantum-gravity tests is the BMV protocol [5,6], which operates at $m \sim 10^{-14}$ kg with coherent position superpositions of order 100 μm and seeks gravitational entanglement rather than field detection. This program is marginally accessible to current technology and is the proper home for the curiosity that motivates single-atom-gravity discussions.

This note is methodological: it documents why a tempting question is not the right question, and identifies where the genuine experimental program lies. The single-atom field is real — it exists, it has a value, it has a sign — but its detection is forbidden by background physics that operates at scales the experiment cannot avoid. The BMV alternative is not just easier; it is the actual target for the underlying physical question.

Acknowledgments

This note belongs to the same imprint as the Confined Photon Cavity (May 2026) and Nano-drum Feasibility (May 2026) standalone notes, sharing their methodology: take a real physics intuition, build the noise budget honestly, identify which limits are fundamental versus engineering, and bound the answer with quantitative numbers. The author thanks the Omega Research Collective red-team reviewers for emphasizing the conceptual hedge — that single-atom field detection would not test quantum gravity even if achieved — which became Section 7 and which is the most important contribution of this note.

Appendix: Numerical Estimates

Quantity	Formula	Value
Hydrogen atom mass	m_H	1.674×10^{-27} kg
Single-atom field at 1 nm	$G m_H / d^2$	1.117×10^{-19} m/s ²
Force on 100 nm test sphere at 1 nm	$G m_H m_{\text{test}} / d^2$	1.117×10^{-37} N
Sota force sensitivity	S_F (Aspelmeyer/Geraci 2021)	$\sim 10^{-20}$ N/ $\sqrt{\text{Hz}}$
Integration time at $d=1$ nm	$(S_F / F)^2$	8.0×10^{33} s
Westphal et al. source mass	$m_{\text{source 2021}}$	90 mg = 5.4×10^{22} atoms
Gap from sota to single-atom	$m_{\text{Westphal}} / m_H$	5.4×10^{22} (22.7 OOM)
Casimir-Polder at 1 nm (atom-sphere)	$3\hbar c \alpha / (8\pi \epsilon_0 d^5)$	$\sim 3 \times 10^{-11}$ N
Patch-potential force at 100 nm	$q_{\text{eff}}^2 / (4\pi \epsilon_0 d^2)$, $V \sim 100$ mV	$\sim 3 \times 10^{-13}$ N
Blackbody pressure at 4 K, 100 nm sphere	$\sigma T^4 A / c$	$\sim 4 \times 10^{-28}$ N
BMV phase, $m=10^{-14}$ kg, $d=200$ μm , $t=1$ s	$G m^2 t / (\hbar d)$	0.32 rad

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